A13482W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3. QUANTUM, ATOMIC AND MOLECULAR PHYSICS

TRINITY TERM 2017

Thursday, 15 June, 2.30 pm – 4.30 pm

Candidates are strongly advised to use the first 10 minutes to read the whole paper before starting writing.

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. Derive an expression for the spin-orbit interaction for an atomic system with a single electron. [*The Thomas precession is a relativistic effect that can be taken into account by making the replacement* $g_s \to g_s - 1$ *for the g-factor of the electron, which can be assumed without justification.*] Show that configurations with $l \neq 0$ are split into two levels separated by

$$\Delta E = \frac{Z^4 \alpha^2 hc R_\infty}{n^3 l(l+1)} \,,$$

where l and n are the orbital angular momentum and principal quantum numbers respectively.

[9]

Calculate the fine-structure splitting of the 2p configuration in atomic hydrogen, expressed in GHz. What is the magnitude of the magnetic flux density, B, experienced by an electron in this configuration? Calculate the splitting of the other electronic configurations belonging to the $n = 2$ and $n = 4$ shells of hydrogen.

[5]

The Balmer-β line in the spectrum of atomic hydrogen corresponds to transitions between the electronic shells with principal quantum numbers $n = 2$ and $n = 4$. For these two shells, draw an energy level diagram (not necessarily to scale) that shows all of the levels labelled with appropriate quantum numbers; indicate any fine-structure splitting and all the allowed electric dipole transitions.

[6]

Calculate the wavelength of the Balmer-β line. For atomic hydrogen, estimate the contribution to the width of this spectral line arising from Doppler broadening in a discharge lamp. Comment on the experimental feasibility of observing the fine-structure splittings.

[5]

[*The Bohr magneton is* $\mu_{\mathrm{B}} = \dfrac{e\hbar}{2m_e} = \dfrac{1}{2}\alpha c e a_0$. *For hydrogenic systems*
$\left\langle \dfrac{1}{r^3} \right\rangle = \dfrac{2}{l\,(l+1)\,(2l+1)} \left(\dfrac{Z}{n a_0} \right)^3$ *where Z is the atomic number.*]

2. Explain what is meant by the LS-coupling scheme, making clear the conditions under which it applies. Illustrate your answer by writing down in spectroscopic notation the terms which arise from the $1s^2 2s^2 2p^6 3s^2 3p\, 4p$ configuration of silicon.

[7]

Show that an atom in the LS-coupling scheme has a g-factor given by

$$g_J \simeq \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} .$$

State the selection rules for L, S, J and M_J in electric dipole transitions in the LS-coupling scheme.

[6]

The ground electronic configuration of mercury is $6s^2$. Three lines are observed in the light emitted from a mercury lamp. A magnetic flux density $B = 1\,\mathrm{T}$ is applied and the lines are observed to have 3, 6 and 9 components respectively, with the frequencies listed in the following table (given relative to $f_A = c/\lambda_A$ and similarly for f_B, f_C).

wavelength/ nm	frequencies of components/ GHz
$\lambda_A = 405$	$f_A, f_A \pm 28$
$\lambda_B = 436$	$f_B \pm 7, f_B \pm 21, f_B \pm 28$
$\lambda_C = 546$	$f_C, f_C \pm 7, f_C \pm 14, f_C \pm 21, f_C \pm 28$

Deduce the values of g_J for the levels involved in lines A and B. All three lines have the same upper term and the same lower term. State the values of L and S for these terms, and the values of J for the levels involved in lines A, B and C respectively, giving your reasoning.

[8]

How accurately does the atomic structure of mercury exhibit the features expected on the basis of the LS-coupling scheme for angular momenta?

[4]

3. The eigenenergies of a diatomic molecule can be written in the form

$$E = E_{\mathrm{e}} + (n + 1/2)\hbar\omega_{\mathrm{v}} + hcBK(K+1)\,.$$

Describe the physical origins of the terms in this equation.

[5]

The selection rules $\Delta n = 1$ and $\Delta K = -1$ gives rise to transitions in the carbon monoxide molecule, CO, with wave numbers: $\tilde{\nu}_{\mathrm{v}} - 3.9, \tilde{\nu}_{\mathrm{v}} - 7.8, \tilde{\nu}_{\mathrm{v}} - 11.7, \ldots\,\mathrm{cm}^{-1}$ where $\tilde{\nu}_{\mathrm{v}} \equiv \omega_{\mathrm{v}}/(2\pi c) = 2143\,\mathrm{cm}^{-1}$. From these data determine the moment of inertia of the molecule.

[5]

Outline briefly why stable diatomic molecules exist and describe the physical interactions that are approximated by the Morse potential of the form

$$V(r) = D\left\{1 - \mathrm{e}^{-\beta(r - r_0)}\right\}^2 \,,$$

where D, β and r_0 are constants. By making suitable approximations find an expression for $\tilde{\nu}_{\mathrm{v}}$ in terms of the parameters of the Morse potential and the reduced mass μ of the molecule.

[9]

A carbon monoxide molecule, $^{12}\mathrm{C}^{16}\mathrm{O}$, has a dissociation energy of $E_{\mathrm{diss}} = 11.16\,\mathrm{eV}$. How is E_{diss} related to the parameters of the Morse potential? From the data in this question, determine the values of β and r_0. Calculate the product βr_0 and comment on its order of magnitude.

[6]

4. Explain why there is a large difference in the threshold pump power required to achieve laser oscillation for three-level and four-level lasers.

[4]

In a laser medium the normalised lineshape function is $g(\omega - \omega_0)$ and ω_0 is the angular frequency of the transition. The population densities of the two levels are N_2 and N_1, and g_2 and g_1 are their statistical weights. Derive an expression for the gain coefficient stating the conditions for which your expression is valid.

[7]

For the case of natural broadening, what is the shape of the gain profile and how does its width depend on the upper and lower state lifetimes? Show that the maximum possible gain cross section is

$$\sigma_{21}(\max) = \frac{2\pi c^2}{\omega_0^2}\,.$$

Evaluate this cross section for a transition at a wavelength of 700 nm.

[5]

In both of the following cases the laser cavity is formed by two mirrors with intensity reflectivities of 100% and 95% at the wavelength of the emitted laser light, and you may assume that $g_2 = g_1$.

a) A 3-level laser has a gain medium in the form of a cylindrical rod of ruby which is 4 cm long and 0.4 cm in diameter. The optically active ions have a concentration of $N_{\text{ions}} = 4 \times 10^{19}\,\text{cm}^{-3}$. The wavelength of the emitted laser light is 694 nm. Estimate the energy of the pumping radiation at 500 nm that must be absorbed by the active ions to achieve laser operation.

b) In a 4-level laser the gain medium is a rod of titanium-doped sapphire that is 1 cm long. The active ions have a concentration of $N_{\text{ions}} = 5 \times 10^{19}\,\text{cm}^{-3}$. The gain cross-section is $\sigma_{21} = 3 \times 10^{-19}\,\text{cm}^2$ for amplification of light in the range $700 - 1000$ nm. The system is pumped by a laser beam at a wavelength of 532 nm with a diameter of 100 μm directed along the rod. The upper laser level has a lifetime of $\tau_2 = 3 \times 10^{-6}$ s and the lower laser level has a lifetime of $\tau_1 = 1 \times 10^{-10}$ s. Estimate the energy that must be absorbed by the active ions for pulsed laser operation. Estimate the minimum power of the pumping radiation required for continuous-wave (cw) operation, explaining your reasoning. Calculate the natural width arising from lifetime broadening and determine whether other line broadening mechanisms are important.

[9]

[*The coefficients in the Einstein treatment of radiation are related by:*

$$A_{21} = \frac{\hbar\omega_0^3}{\pi^2 c^3} B_{21} \text{ and } g_1 B_{12} = g_2 B_{21}\,. $$]